Data Flow Diagram

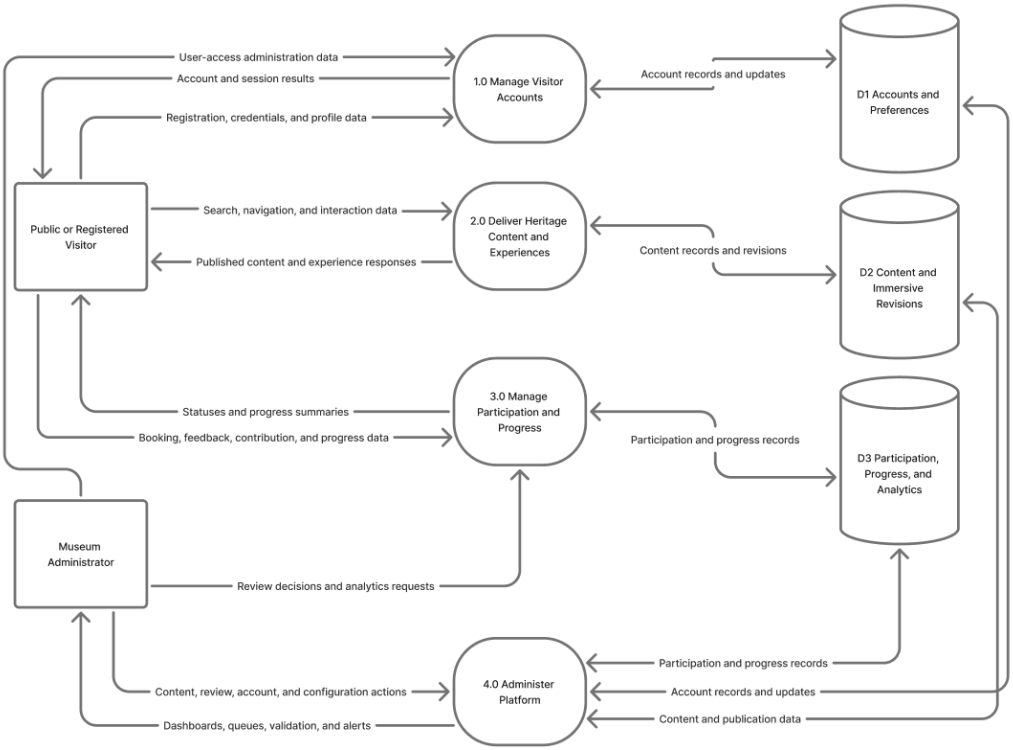


Figure A47. Level 1 data flow diagram of the WebXR-Enabled Digital Heritage Platform.

Figure A47 summarizes how account access, heritage experiences, participation and progress,and platform administration transform visitor and administrator data and read or update the system's account, content, and operational data stores.

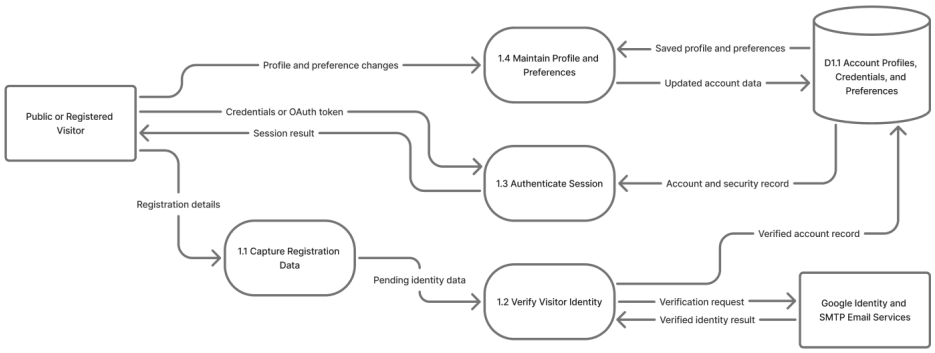


Figure A48. Visitor registration, identity verification, authentication, and profile data flow.

Figure A48 shows how visitor account details are validated through email or Google identity services, stored as account records, used to create an authenticated session, and updated when profile or preference data changes.

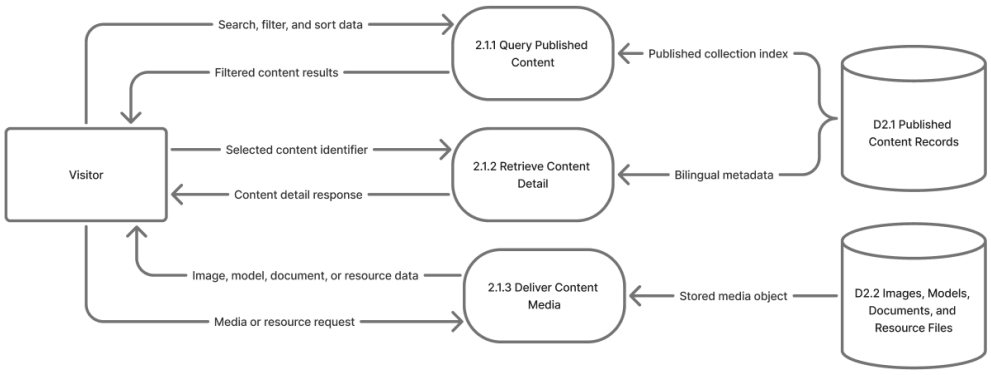


Figure A49. Online Gallery and Educational Resources discovery data flow.

Figure A49 shows how visitor search and selection data are transformed into published collection results, bilingual content details, and the corresponding image, model, document, or learning-resource media.

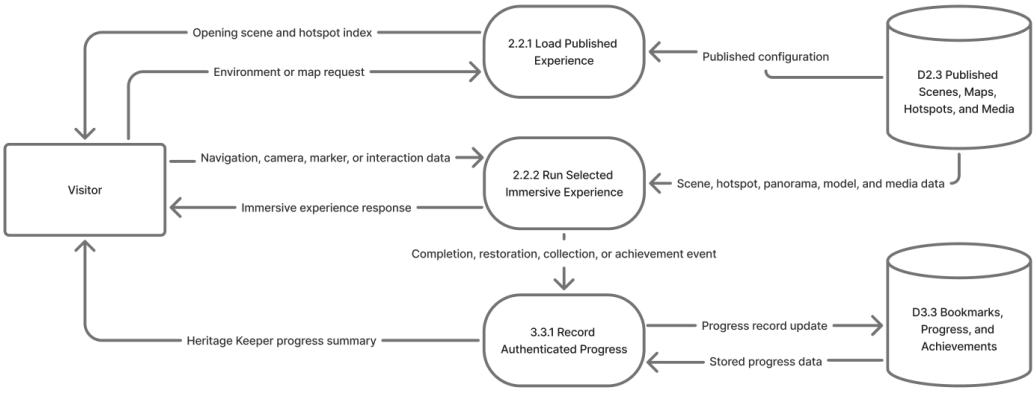


Figure A50. Virtual Tour, Site Map, Heritage Trail, augmented-reality, and WebXR data flow.

Figure 50 shows how visitor environment and interaction data retrieve published scenes, map configuration, hotspots, media, and models, while authenticated completion and achievement events are transferred to the progress store.

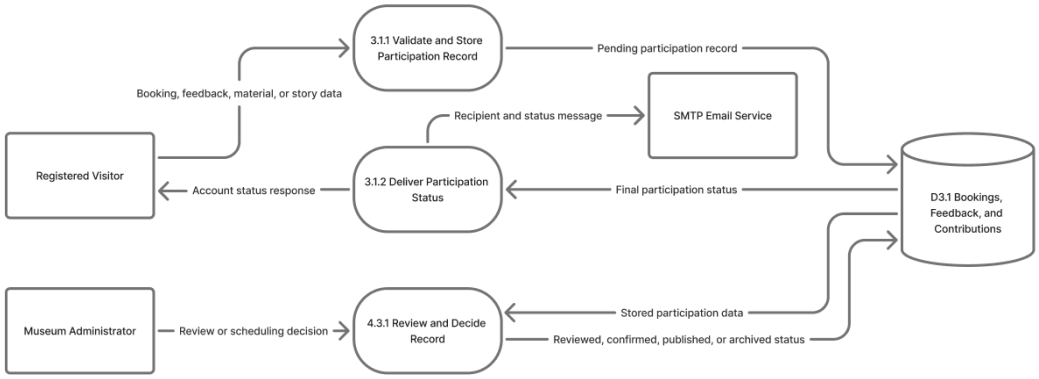


Figure A51. Guided-visit booking, feedback, contribution, review, and notification data flow.

Figure A51 shows how authenticated participation data become stored pending records, how administrators retrieve and decide those records, and how the system returns account or email status information to the registered visitor.

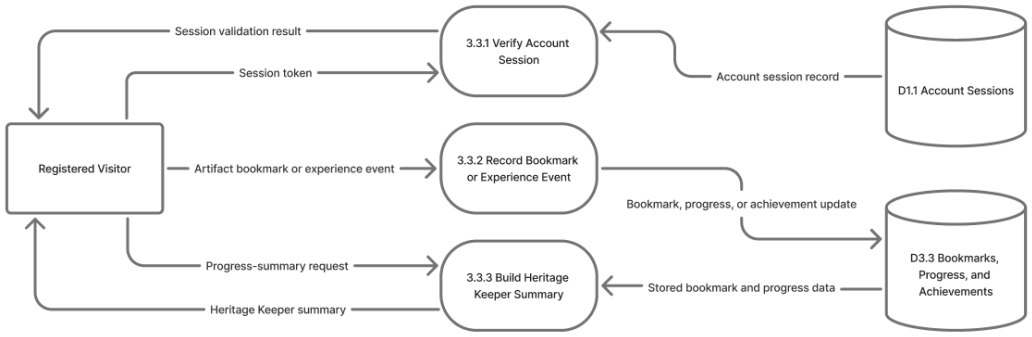


Figure A52. Bookmark, immersive-progress, achievement, and Heritage Keeper data flow.

Figure A52 shows how the registered visitor's session is verified before bookmark or experience data are stored and how the consolidated bookmark, restoration, trail, WebXR, and achievement records are returned as a Heritage Keeper summary.

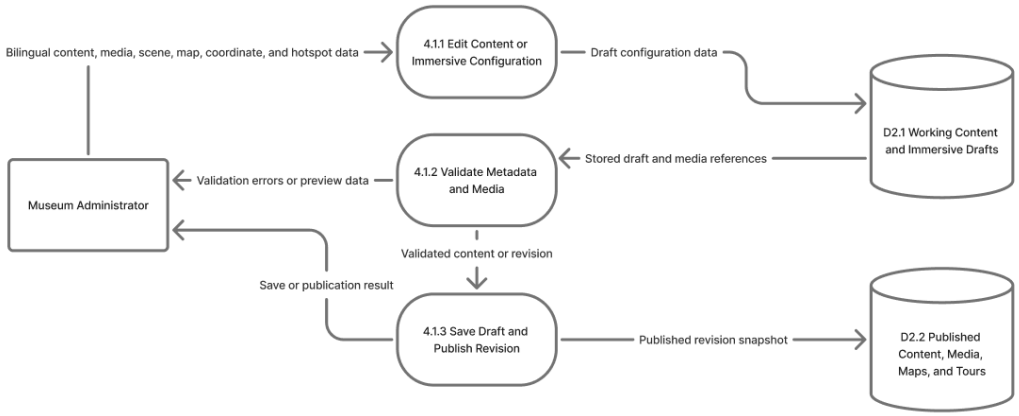


Figure A53. Public and immersive content editing, validation, revision, and publication data flow.

Figure A53 shows how administrator-supplied bilingual metadata, media, scenes, maps,coordinates, and hotspots are validated, stored in a working draft, and promoted to the public content and immersive-revision store after a successful publish check.

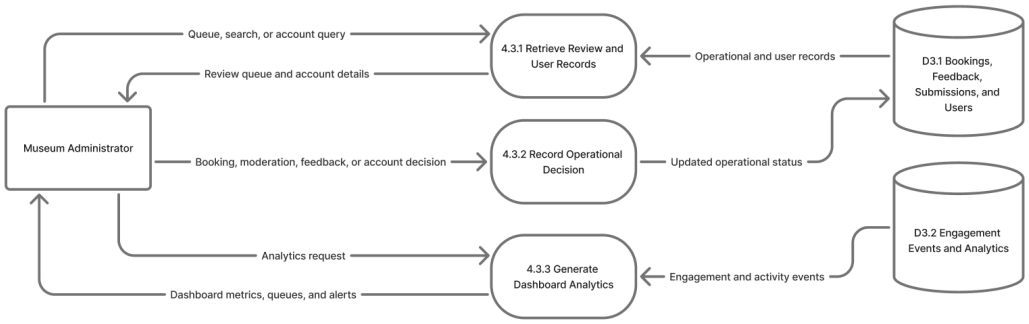


Figure A54. Administrative review, public-user oversight, and analytics dataflow.

Figure A54 shows how the administrator requests operational queues and analytics, receives booking, submission, feedback, account, and engagement data, records review or access decisions, and receives updated dashboard and notification results.

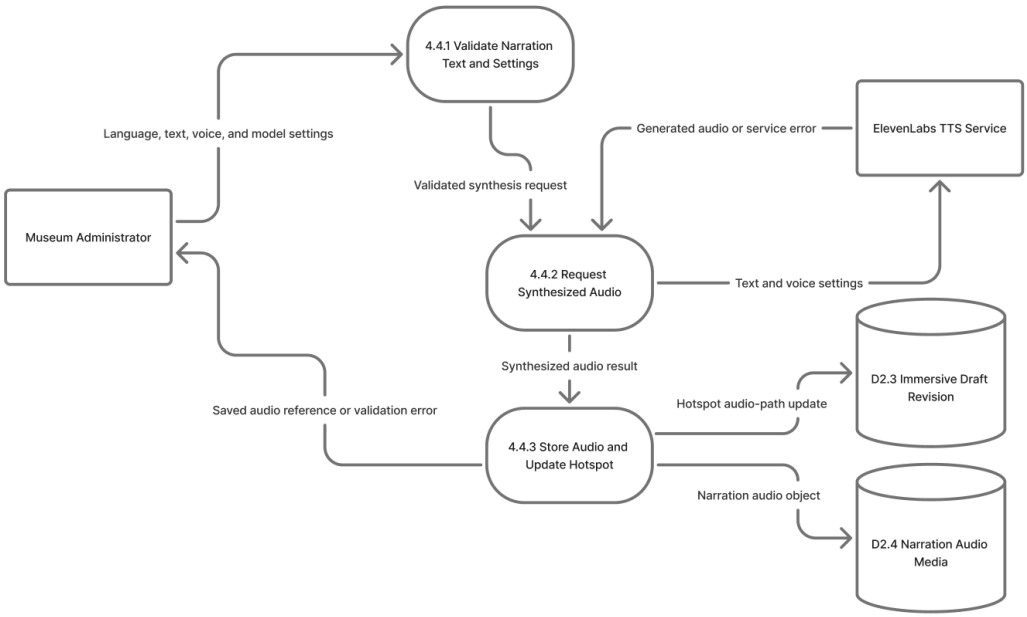


Figure A55. Text-to-speech narration generation and storage data flow.

Figure A55 shows how narration text, language, voice, and model settings are validated and sent to ElevenLabs and how the generated audio object and path are stored in the working immersive revision for later publication.